

Online Retail Dataset using HBase & HiveQL

◆ i. Create and Load OnlineRetail Table in Hive

Start Hive:

```
Bash
hive
```

Create Table

```
SQL
CREATE TABLE online_retail(
invoice_no STRING,
stock_code STRING,
description STRING,
quantity INT,
invoice_date STRING,
unit_price FLOAT,
customer_id STRING,
country STRING
)
ROW FORMAT DELIMITED
FIELDS TERMINATED BY ',';
```

Load Data from Local File

```
SQL
LOAD DATA LOCAL INPATH '/home/cloudera/online_retail.csv'
INTO TABLE online_retail;
```

Display Data

```
SQL
SELECT * FROM online_retail LIMIT 10;
```

◆ j. Create Index on Online Retail Table

```
SQL
CREATE INDEX idx_country
ON TABLE online_retail(country)
```

```
AS 'COMPACT'  
WITH DEFERRED REBUILD;
```

Rebuild Index

```
SQL  
ALTER INDEX idx_country ON online_retail REBUILD;
```

◆ k. Find Total and Average Sales

Total Sales

```
SQL  
SELECT SUM(quantity * unit_price) AS total_sales  
FROM online_retail;
```

Average Sales

```
SQL  
SELECT AVG(quantity * unit_price) AS average_sales  
FROM online_retail;
```

◆ l. Find Order Details with Maximum Cost

```
SQL  
SELECT invoice_no,  
       description,  
       quantity,  
       unit_price,  
       (quantity * unit_price) AS total_cost  
FROM online_retail  
ORDER BY total_cost DESC  
LIMIT 1;
```

◆ m. Find Customer with Maximum Order Total

```
SQL  
SELECT customer_id,  
       SUM(quantity * unit_price) AS total_order  
FROM online_retail  
GROUP BY customer_id  
ORDER BY total_order DESC  
LIMIT 1;
```

◆ n. Find Country with Maximum and Minimum Sale

Maximum Sale Country

```
SQL
SELECT country,
       SUM(quantity * unit_price) AS total_sales
FROM online_retail
GROUP BY country
ORDER BY total_sales DESC
LIMIT 1;
```

Minimum Sale Country

```
SQL
SELECT country,
       SUM(quantity * unit_price) AS total_sales
FROM online_retail
GROUP BY country
ORDER BY total_sales ASC
LIMIT 1;
```

◆ o. Create External Hive Table Connected to HBase

Start HBase:

```
Bash
hbase shell
```

Create HBase Table

```
Bash
create 'online_hbase', 'info'
```

Insert Sample Data

```
Bash
put 'online_hbase', '1', 'info:invoice', '1001'
put 'online_hbase', '1', 'info:item', 'Laptop'
put 'online_hbase', '1', 'info:country', 'India'

put 'online_hbase', '2', 'info:invoice', '1002'
put 'online_hbase', '2', 'info:item', 'Mouse'
put 'online_hbase', '2', 'info:country', 'USA'
```

Open Hive:

```
Bash
hive
```

Create External Hive Table

```
SQL
CREATE EXTERNAL TABLE online_hive(
id STRING,
invoice STRING,
item STRING,
country STRING
)

STORED BY 'org.apache.hadoop.hive.hbase.HBaseStorageHandler'

WITH SERDEPROPERTIES (
"hbase.columns.mapping"=":key,info:invoice,info:item,info:country"
)

TBLPROPERTIES ("hbase.table.name"="online_hbase");
```

◆ p. Display Records of OnlineRetail Table in HBase

```
Bash
scan 'online_hbase'
```



Useful Commands

Show Hive Tables

```
SQL
show tables;
```

Show HBase Tables

```
Bash
list
```

Describe HBase Table

```
Bash
describe 'online_hbase'
```

Exit Hive

```
SQL
quit;
```

Exit HBase

```
Bash
exit
```

